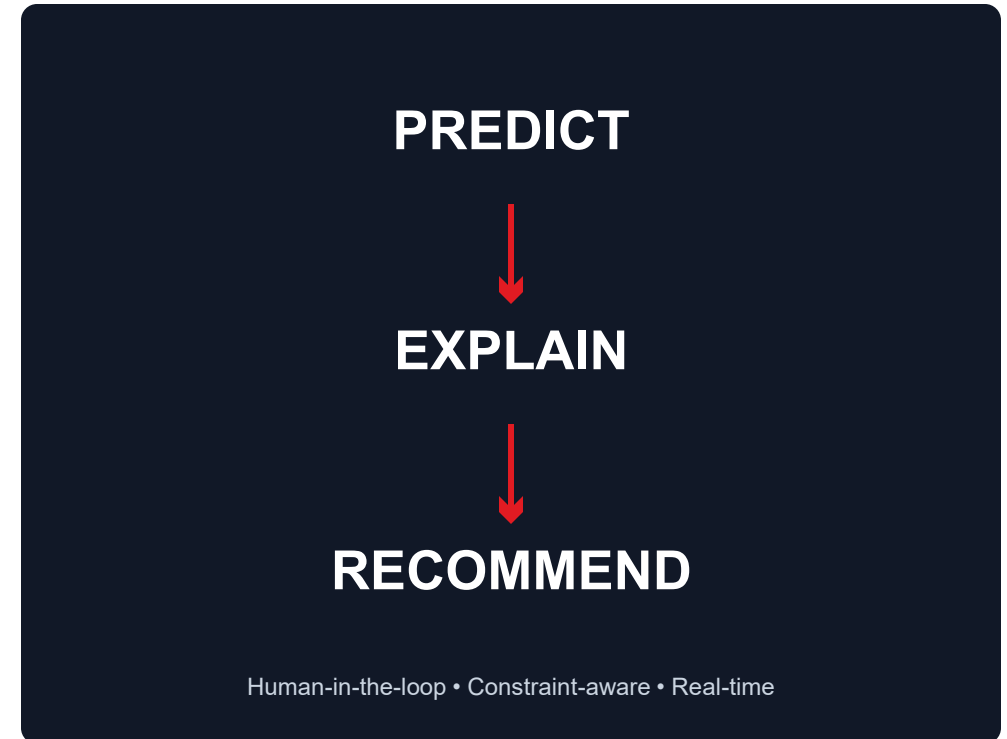


GRADESENSE

Explainable AI for safer, faster paper grade transitions

- **Honeywell Hackathon Final Submission**
- **Industrial AI | Software Prototype**
- **Bhavuk Mahajan | GradeSense**



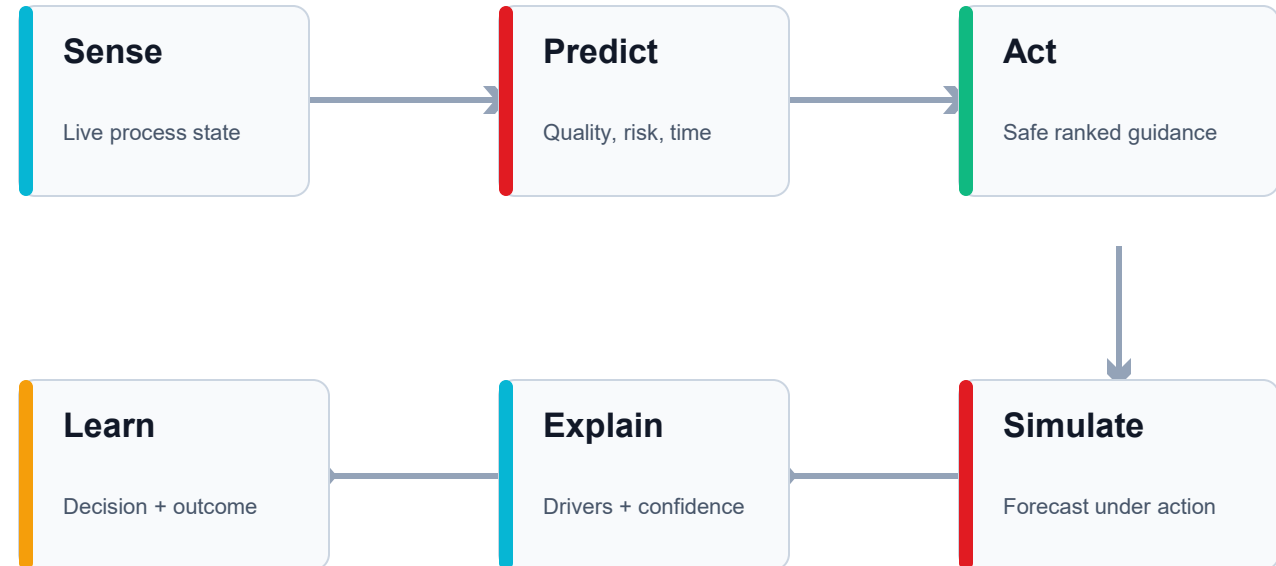
PREDICT BEFORE QUALITY DRIFTS

THE PROBLEM

Grade transitions are multivariable, time-dependent, and costly when deviation is detected late.

Operators must connect changing sensor values, future basis-weight behavior, safety constraints, and past outcomes under time pressure.

THE GRADE SENSE LOOP



One governed system connects detection, explanation, intervention, and measurable learning.

WHY GRADESENSE IS DIFFERENT

WHY THIS APPROACH

Forecast the whole trajectory, then simulate only constraint-valid actions with the same active model.

The result is earlier warning without recursive error accumulation - plus recommendations an operator can verify before acting.

DESIGN ADVANTAGES

- Windows never cross transitions
- Empirical confidence bands
- Safety gate before inference
- Human approval + outcome audit

APPROACH COMPARISON

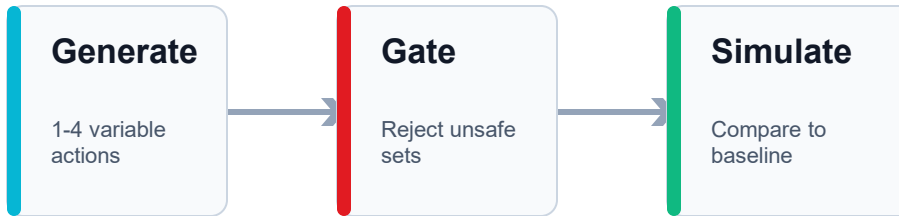
METHOD	WARNING	ACTION QUALITY	TRUST
Reactive thresholds	Late	None	Low
Recursive forecast	Early	Error compounds	Medium
Black-box optimizer	Early	Needs external guardrails	Low
GradeSense	Early + 12-step	Pre-checked + simulated	High

EXPLAINABILITY, SAFETY, AND OUTCOMES

EXPLAINABLE AI



RECOMMENDATION ENGINE



**Only positive, constraint-valid improvements
are ranked for the operator.**

BUSINESS IMPACT

EARLIER

risk detection

SAFER

set-point guidance

TRACEABLE

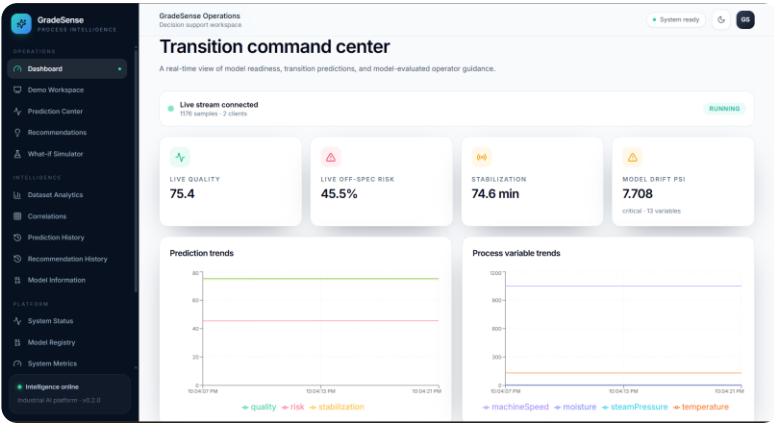
decision-to-outcome loop

Advisory by design: the operator remains in control.

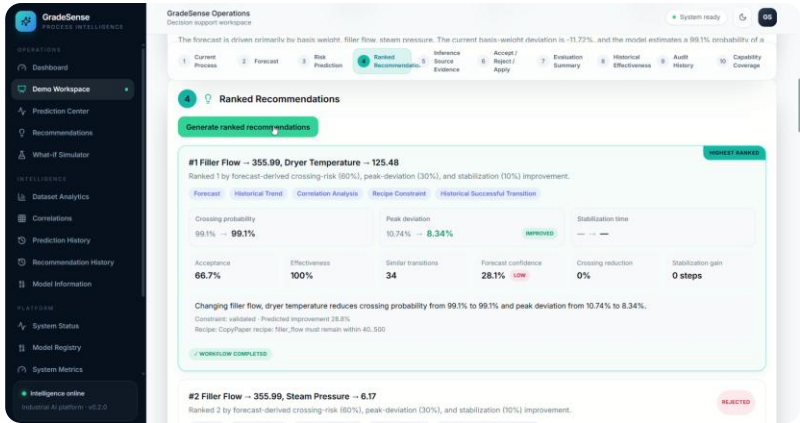
VALIDATED ANALYTICS + WORKING PRODUCT



Validation is grouped by complete transitions: 192 train / 48 held out. Windows never cross transition boundaries.



REAL-TIME RISK + DRIFT



RANKED, EXPLAINED ACTIONS

TRY IT, REVIEW IT, BUILD ON IT

GradeSense turns model output into safe, explainable, measurable operator action.

LIVE PRODUCT

<https://gradesense-4weh.onrender.com/>

SOURCE CODE

<https://github.com/bhavukm007/GradeSense>

API DOCUMENTATION

<https://gradesense-api.onrender.com/docs>

DEMO VIDEO

<https://drive.google.com/file/d/1rCr3CzuCit0O029VzyxIhHoR1lv9E93E/view?usp=sharing>

Pilot next: historian integration, mill-specific calibration, identity controls, and measured avoided off-spec production.